

The Acquisition of Pointing Signs and Pronominal Systems in Sign Languages

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Abstract

In American Sign Language (ASL), pronominal functions are carried out by pointing signs: points to self for first-person, points to addressee or non-addressed persons or things as second- and third-person. This picture is complicated by the fact that pointing signs are also used in other ways in ASL: as one example, a signer can point to an abstract locus to refer to a non-present referent associated with that locus.

Analysis of these pointing signs has been controversial, starting with arguments that points are not really linguistic elements, just deictics (Johnston 2013). Even when considered as linguistic, there are numerous debates over which analysis is the most appropriate. For example, Meier (1990) argues that the ASL pronominal system does not encode any grammatical distinction between 2nd and 3rd person, only a division between 1st and non-1st person.

We provide evidence from the acquisition of ASL and from co-speech pointing in young hearing children to contribute to the discussion of how pointing signs should be analyzed. The acquisition pattern that emerges shows first that the distribution of points to objects versus people is very different. Both signing and non-signing children produce points to objects from an early age. However, only signing children develop the use of points to self and other people before 3;00, highlighting a difference between gestural and linguistic pointing. Furthermore, the development of points to different persons supports the linguistic analysis, and provides potential support for the 2nd/3rd-person distinction. On the other hand, points to abstract loci are not used by either group of young children, suggesting that this is yet another kind of element.

1. Introduction

Pointing is ubiquitous – it is used in various forms all over the world, by deaf and hearing people of all ages, for the purpose of picking out both concrete objects and locations, as well as metaphorical or abstract referents (Kita, 2003). Pointing can be used independently, or accompany spoken language as a co-speech gesture. Pointing behavior emerges in young children quite early (Masataka, 2003), and when produced in combination with lexical items, serves as a harbinger for the imminent development of two-word utterances (Goldin-Meadow & Butcher, 2003; Iverson & Goldin-Meadow, 2005). Clearly, pointing is a vital part of the multimodality of language (Müller et al., 2013, 2014).

It should come as no surprise then that pointing is an important component of sign language grammars. Pointing, whether with the index finger, another handshape, or by eye gaze and body position – serves multiple functions in sign language grammars (Cooperrider & Mesh, 2022; Cormier, 2012). Here we will focus on index finger pointing to people, objects, and locations. These points (or perhaps a subset of them) are frequently analyzed as pronouns in sign languages, but such analyses are fraught with controversy due to numerous ways in which sign language pointing appears to behave differently from pronouns in spoken languages (see section 2). In this paper, we will address some of the issues that have been raised regarding the analysis of pointing as pronominals in sign languages, through the lens of language development. We will suggest that acquisition data brings useful insights to the debate over the how to best analyze pointing / pronouns in sign linguistics.

The chapter begins with an overview of some proposals regarding the analysis of pronouns in sign languages (with a focus on ASL), and discusses questions that these proposals raise. We bring up a specific theoretical hypothesis about the organization of pronouns, which is

supported by data from the acquisition of spoken languages. This sets the stage for the current study. Here, we examine the patterns of pointing used by hearing, non-signing children, and contrast them with the patterns observed in our study of four children acquiring ASL from their deaf, signing parents. We conclude the chapter with some suggestions for future research.

2. Pointing and Pronouns in Sign Languages

For over half a century it has been recognized that signers of ASL and other sign languages use points to self, addressee, non-addressed persons, objects, and locations in ways that appear to function analogously to personal, demonstrative, and locative pronouns (Cormier, 2012; Friedman, 1975; Meier & Lillo-Martin, 2010, 2013; among many others). For example, to refer to self, signers of ASL and many other sign languages use an index finger point to their own chest. To refer to an addressee or a non-addressed person, signers point toward that person's chest. These pointing signs can be used in isolation, but they are also used in sentences, with a distribution that is overall very similar to that of pronominals in spoken languages.

Given the apparent pronominal function of (at least some) pointing signs, various researchers have proposed that they should be analyzed in essentially the same way as personal pronouns in spoken languages (Kegl, 1987, 2003 (originally distributed in 1976); Lillo-Martin & Klima, 1990). However, others have argued that a pronominal analysis ignores the fact that pointing signs overlap considerably with co-speech gestural pointing in form and function, and display characteristics that are quite distinct from spoken language pronouns (Ahlgren, 1990; Friedman, 1975; Johnston, 2013). Johnston's corpus study of Auslan (Australian Sign Language) found considerable variability in the forms of pointing signs that were used referentially, along with cases where pointing could be interpreted as serving more than one function (e.g., picking

out both a referent and its location). Such overlapping function is not observed among pronoun forms in spoken languages.

Another characteristic of pointing signs that distinguishes them from spoken pronouns is the fact that the form of the sign is not categorical, particularly for signs not directed at the self. Here we put aside variations in handshape (such as those observed by Johnston) and focus on the fact that the pointing sign is directed toward the location of the referent, wherever that referent happens to be. Consequently, it is impossible for a sign language lexicon to list all possible (non-first) pronouns with all of their specifications for movement and location. Therefore, the putative lexical entry for a pronominal pointing sign must indicate that the index finger is directed toward the *location of the referent*, something that is not normally considered to be a linguistic element (Liddell, 2000). Furthermore, because an addressee and a non-addressed referent may be located anywhere in the vicinity of a signer, it is not possible to distinguish the form of a pointing sign that picks out an addressee (2nd person) versus a non-addressed (3rd person) referent (Meier, 1990). Some researchers have used the concurrence of eye gaze with the direction of the point to indicate 2nd person, while discontinuity in eye gaze and point direction would indicate 3rd person (e.g., Berenz, 2002). However, in an eye-tracking study of ASL signers, Thompson et al. (2013) did not find evidence to support that claim.

The considerations outlined above have led some researchers (e.g., Cormier et al., 2013; Fenlon et al., 2019; Liddell, 2000; Meier & Lillo-Martin, 2010) to propose various analyses that treat pointing signs as a combination of linguistic elements (i.e. the manual form of the pointing sign) and gestural elements (i.e., the location toward which the pointing sign is directed). Under (some of) such analyses, a pointing sign (glossed IX, for ‘index’) is treated on par with the combination of a spoken language pronoun and a co-speech gestural point. Furthermore, some

analyses consider the formal person-marking distinction in sign languages to be limited to 1st vs. non-1st person (e.g., Engberg-Pedersen, 1993; Lillo-Martin & Meier, 2011; Meier, 1990), while others argue that sign languages maintain a full 3-way person distinction (e.g., Alibašić Ciciliani & Wilbur, 2006; Barberà Altimira, 2015).

The description so far has been limited to points directed at referents in the physical space surrounding a signer. However, when signers wish to pick out a non-present referent, pointing signs may be used in this context as well. Signers are said to ‘assign’ a referent to a locus – a location in the space around them which is abstractly associated with the referent for the purposes of establishing a locus towards which pointing signs can be directed (Lillo-Martin & Klima, 1990; Padden, 1983). Taking into consideration the behavior of signs directed towards these abstract loci, some researchers have proposed analyzing such pointing signs as demonstrative pronouns (Ahlgren, 1990; Koulidobrova & Lillo-Martin, 2016). Another proposal is that pointing signs directed at loci should be analyzed as modifiers, whose meaning is identical to that of co-speech pointing gestures; roughly, an IX to locus A means ‘there is an x which is located at A’ (Ahn, 2020). Without going into more detail, we want to allow for the possibility that pointing signs directed at abstract loci might have a different status from those directed at present referents.

Based on the discussion so far, we propose to evaluate a possible model of the pronominal system of ASL grounded in the feature geometry for pronouns proposed by Harley & Ritter (2002). Their proposal is based on morphosyntactic properties of pronouns found across a wide variety of languages, including systems observed in a database of 110 languages. Harley & Ritter also provide acquisitional evidence in support of their proposal (pulled from 10 previous acquisition studies). The hierarchical model they propose is given in Figure 1 below.

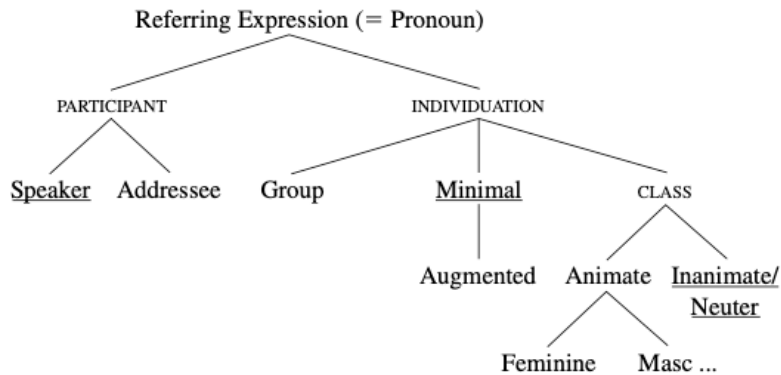


Figure 1. Feature geometry model for pronouns proposed by Harley & Ritter (2002).

In this model, features are monovalent (as opposed to common binary features), so they only appear in the hierarchy for a particular pronoun when they apply. The hierarchical structure of the model reflects dependency implications (e.g., a system will make a gender distinction only if it makes an animacy distinction). The more nodes that dominate a feature, the more marked that feature is (e.g., the animacy feature is more marked than the participant feature). Finally, the underlined features in Figure 1 are default values, predicted by Harley & Ritter to be acquired earlier than the non-default forms within each organizing node; however, (the default forms for) each node might not always be acquired in the same order.

Given the discussion presented earlier, we will use acquisition data to test the applicability of this model for IX signs in ASL, taking into consideration some of the theoretical questions that emerge about these signs in the literature. Our research questions are as follows.

Research questions

- A. **How comparable are pointing signs and co-speech points?** In terms of first language acquisition, are the developmental paths associated with pointing signs (for signing

children) and (potentially) co-speech points (for non-signing children) similar, or are there substantial differences which might indicate analytic distinctions?

- B. **Do pointing signs behave like pronouns?** In terms of first language acquisition, do we see a similar developmental pattern for the different uses of pointing signs mirroring developmental patterns found for the acquisition of pronouns in spoken languages?
- C. **Is there evidence for a distinction between 2nd and 3rd-person pointing signs?** Does the time course of acquisition provide evidence for a developmental distinction between 2nd and 3rd person pointing signs in ASL?
- D. **Should pointing signs directed to abstract loci be analyzed differently from other pointing signs?** Do signing children acquire these two types of pointing differently?

Before turning to the two studies we conducted to address these questions, we briefly summarize previous research on the development of pointing by non-signing (section 3) and signing (section 4) children.

3. Previous Research on the Development of Pointing by Non-Signing Children

It has frequently been observed that non-signing hearing children begin to use pointing at a very early age, typically between 9 and 12 months (Bates, 1976). There is also much evidence that children's early use of pointing is related to their later language development (see Colonna et al., 2010 for a meta-analysis). However, almost all of the extensive research on pointing in non-signing children concerns points to objects and/or locations, with very little discussion of children's use of pointing to self, addressee, or other persons.

A study by Pizzuto & Capobianco (2008) examined monthly video recordings of seven children acquiring Italian, between the ages of 12 and 24 months. This study found that all of the children pointed to objects and locations from the very first recording, but points to people—particularly points to self and addressee – were only found for some of the children, and those who produced them used them at a very low frequency, and only from the age of 16 months or later. Thus, the development of pointing in non-signing children may be very different from that of pointing used by signing children, an issue we raised in our first research question.

4. Previous Research on the Development of Pointing Signs in Sign Languages

The first systematic examination of pointing development by signing children was conducted by Petitto (1987), in a study that examined data from two deaf children ages 6 months to 2;03 acquiring ASL from their deaf, signing parents. Petitto reported that both children productively pointed to objects and locations by the age of 12 months, similarly to the Italian children described by Pizzuto and Capobianco. However, Petitto also reported that by this age, both signing children also pointed toward other people, and one child pointed toward herself at this age. Strikingly, both children then reportedly stopped pointing to people (either to themselves or to addressees) between the ages of 12 and 18 months, although they continued to point to objects and locations during this time. Furthermore, Petitto found that when points to self and addressee resumed (ages 21-22 months), the children also produced pronoun reversal errors (one child consistently and the other inconsistently). These reversal errors are similar to errors that have been observed for some hearing children acquiring spoken language pronouns (see Clark, 1978), but are surprising for ASL, given the transparent mapping of pronoun points and their referents in sign languages. Petitto argues that these errors are incompatible with a view

that considers pointing in sign languages as non-linguistic; further, she claims that when signing children stop pointing to themselves and addressees, then start again, but with pronoun reversal errors, this sequence indicates that children acquiring ASL initially analyze pointing as gestural forms, but eventually reanalyze them (not without error) as pronominal forms.

Petitto's study has been highly influential, but it should be noted that the data come from only two participants, with 12 one-hour sessions each. Two additional studies have been published on the development of pointing in children acquiring sign languages other than ASL, although they are also based on very limited data sets. Hatzopoulou (2011) studied one deaf child acquiring Greek Sign Language between the ages of 12 and 27 months. Like Petitto, Hatzopoulou found early use of points to objects/locations, with some points to people (including self) at 14-15 months. Although the rate of this child's pointing to people never dropped to zero in subsequent months, it decreased sharply between 16-20 months to only 1% of all points that she produced. Pointing to people resumed a higher proportion of usage after 20 months.

Morgenstern and colleagues (2010, 2016) studied two children simultaneously acquiring French Sign Language (LSF) and spoken French. One child was deaf, and one was a hearing child of deaf parents (coda); they also studied one hearing non-signing French-acquiring child. The deaf child produced points towards people throughout the observation period (0;08-2;00); however, points to self dropped out completely between 0;11 and 1;03. Similarly, the bimodal bilingual child produced points to self at 1;01 and 1;02, but then none until 1;07. The hearing, non-signing child did not produce any points to self during the observation period, similar to the hearing Italian children observed by Pizzuto and Capobianco (2008).

These three studies of signing children were based on small data sets, and the details of the children's development differed in various respects. Nevertheless, a remarkably consistent

overall developmental sequence can be detected. Points to objects and locations are frequent throughout the observation period, extending from before the age of 12 months until approximately 24 months. At a very early age, points to self (and in some cases, other people) may be observed, but these then drop out for a period of time, appearing again only after several months. In contrast, the limited data available from hearing, non-signing children suggest that over the same age span, they produce very few (if any) points to themselves and others. While the consistency of this general pattern among signing children and the contrast between their early pointing behaviors and those of non-signing children are striking, these remain largely descriptive patterns that do not in themselves tell us much about how signing children develop their pronominal system. We turn now to two studies we conducted to address these empirical gaps, the first examining non-signing children, and the second deaf children from deaf, signing parents. Together, these studies provide data necessary for considering our broader research questions about the status of pointing signs and pronouns in sign languages.

5. Study 1: Hearing, Non-Signing English-Speaking Children

5.1 Participants and Data Set

We examined index-finger pointing produced by four hearing, English-speaking children, ages 1;06-4;00. We used data from Alex, Lily, Naima, and Violet from the Providence Corpus (Demuth et al., 2006; Demuth & McCullough, 2009) of the CHILDES database (MacWhinney, 2000), as this corpus includes video recordings of children in the appropriate age range. The recordings are naturalistic language samples with each child and their parent, and are transcribed using the CHAT format (MacWhinney, 2025).

5.2 Coding and Analysis

Video recordings were viewed and all points using the index finger were noted. Coding included points that culminate in a touch to an object or person; however, points to indicate body parts (e.g., in response to, “Where’s your nose?”) were excluded. Actions of reaching or manipulation were also excluded.

Points were categorized into three types, as follows:

Points to Objects/Locations	Points to present objects and/or locations
Points to Self	Points to the child themselves, usually occurring at chest level
Points to Other People	Points to people in the environment other than self (usually, this is the addressee)

5.3 Results

The total number of points produced, number of points to objects/locations, and number of points to people are given in Table 1, as well as the age at which points to self or other people were observed. Note that in some cases, children were engaged in an identification game in which the parent asked, “Where’s X”, and the child responded by pointing to X (e.g., “Where’s Mama?” elicited points to the child’s mother). Table 1 reports both the total instances of pointing to people *including* those elicited in identification games, as well as the total *excluding* those instances (in parentheses).

Table 1. Points produced by hearing, non-signing, English-speaking children.

Child	Age	#Ses-	Time	Total	#Points to	#Points to	First	First Points
	Range	sions	[hh:mm]	#Points	Object/	Person*	Points	to Other
	(y;m)				Location		to Self	Person
Alex	1;06-3;06	9	8:58	528	526	2 (0)	NYO	NYO
Lily	1;06-4;00	10	9:34	459	457	2 (2)	3;03	3;03
Naima	1;06-3;10	10	10:43	238	223	10 (8)	NYO	3;00
Violet	1;06-4;00	10	7:13	294	291	3 (1)	NYO	4;00

Notes:

* Total number of points to persons; (number of points to person excluding ‘where’s X’ games).

NYO = not yet observed outside of ‘where’s X’ games.

5.4 Discussion

These results replicated earlier findings that points to objects and locations are highly productive for hearing, non-signing children, while points to people are extremely rare. Points to self and other people are not observed before age 3.

Analysis of the English transcripts for these hearing children revealed that all four used spoken 1st and 2nd person pronouns by age 2;00-2;03. Apparently, in contrast to the many ways in which pointing supports the development of spoken words (see Section 3), it is not the case that pointing to people is a prerequisite for the development of personal pronouns in spoken English.

6. Study 2: Deaf Children with Deaf, Signing Parents

6.1 Participants and Data Set

Our study of the development of pointing signs by deaf children with deaf, signing parents¹ makes use of data from a longitudinal study of four children (Lillo-Martin & Chen Pichler, 2006, 2008), recorded in naturalistic play sessions with their parent(s) and/or signing research assistants (both deaf and hearing signers). The age ranges and other information about the sessions we analyzed are provided in Table 2 below.

Table 2. Data set: Deaf children with deaf parents.

Child	Age Range	# Sessions	Time	Total IX
	[years;months]		[hours:minutes]	
ABY	1;05-3;00	30	27:12	2295
JIL	1;07-3;00	25	19:20	1338
NED	1;06-2;11	26	22:10	1576
SAL	1;07-2;10	18	16:55	2769

Prior to our coding, the video recordings were annotated in ELAN (Crasborn & Sloetjes, 2008), following the conventions of the SLAASh (Sign Language Acquisition, Annotation, Archiving and Sharing) project (Hochgesang, 2022). Under those conventions, all index finger pointing signs are glossed as IX; points to self are glossed IX_1, and points to multiple locations

¹ NED has one deaf parent (mother) and a hearing parent (father) who is a fluent signer; the other three have two deaf, signing parents.

as IXarc. Additional information, such as the referent person/object/location pointed to, is provided on an ‘appendix’ tier.

6.2 Coding and Analysis

All coding and analyses were conducted by viewing the video recordings together with the ELAN annotation files. We considered all instances of children’s pointing, and classified the referent of the point as follows. Index finger pointing was coded regardless of the presence or absence of other signs in an utterance.

1st	Points to the self, usually occurring at chest level
2nd	Points to the addressee; addressee is contextually determined as the person to whom the child’s utterance is directed
3rd	Points to a person in the environment who is not the person to whom the child’s utterance is directed
Inanimate	Points to an object/location

Our analysis focuses on the age at which a type of sign can be considered ‘acquired’. Following Snyder (2007), we determined the age of First of Repeated Uses (FRU) by locating the first instance of a category followed by another use in the next month. In order to determine whether any two categories are acquired at significantly different ages, we used the Binomial test, which takes into consideration relative frequency of the compared items at a later age.

6.3 Results

Across the four children, we observed different absolute ages for acquisition of the different types of points, but consistency in terms of the acquisition sequence, as follows.

1. IX to inanimate objects/locations is always present from the first observation, for all four children.
2. IX to 1st person is either present from the first observation or is acquired earlier than (or at the same time as) IX to any other animate referent.
3. IX to 2nd and 3rd person appear later; 3rd person IX is very infrequent in the data.
4. IX to plurals referents generally occurs later still, and is also infrequent.

Figure 2 provides a visual representation of the ages of acquisition of the four pointing types across the four children. An ellipse across categories indicates no significant difference in the age of appearance (FRU); separated ellipses indicate significant differences. The statistical results are provided in Table 3.

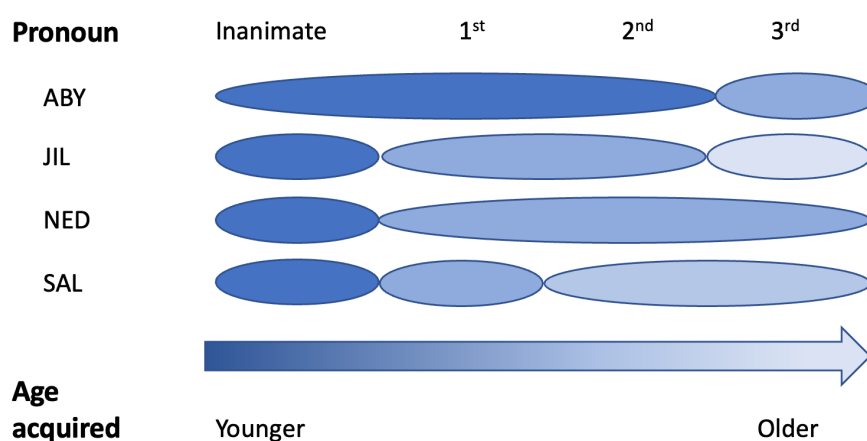


Figure 2. Visual representation of acquisition sequence

Table 3. Statistical comparisons

	FRU IX	FRU IX_1	FRU IX(addr)	FRU IX(non-addr)	p(IX vs. IX_1)	p(IX_1 vs. IX(addr))	p(IX(addr) vs. IX(non-addr))
ABY	1;05	1;05	1;05	1;07	(N/A)	(N/A)	0.003
JIL	1;07	1;10	1;10	2;00	0.0275	(N/A)	0.028
NED	1;06	2;00	2;03	2;03	<.001	0.56	(N/A)
SAL	1;07	1;08	1;11	2;02	0.008	<.001	0.132

7. General Discussion

Let us refer to the four research questions raised earlier for the discussion of our results.

- A. **How comparable are pointing signs and co-speech points?** In terms of first language acquisition, are the developmental paths associated with pointing signs (for signing children) and co-speech points (for non-signing children) similar, or are there substantial differences which might indicate analytic distinctions?

Recall from our study of four children acquiring English between the ages of 1;06-4;00 that none of the children used points to self or other people before the age of 3, and even then, the overall number of such points (outside of ‘where’s X’ games) was extremely low. Pizzuto & Capobianco (2008) also found very limited use of points to self and addressee by Italian-acquiring children before the age of 2; we do not know of other evidence regarding pointing to people by young non-signing children.

In contrast to the non-signing children, the four signing children in our study all used points to people in the period of study. For all four children, the FRU criterion was met for self-

reference by age 2 (range 1;05-2;00), and for reference to addressee at the same age or shortly thereafter (range 1;05-2;03). There were fewer opportunities for points to non-addressed persons, as the filming sessions frequently included only the child and one other person on-screen; however, all four participants achieved FRU for non-addressed referents (range 1;07-2;03). These results reconfirm that pointing is acquired very differently for signing and non-signing children. But does the acquisition pattern support an analysis of pointing by signing children as pronouns? This is the issue addressed by our next research question.

B. Do pointing signs behave like pronouns? In terms of first language acquisition, do we see a similar developmental pattern for the different uses of pointing signs mirroring developmental patterns found for the acquisition of pronouns in spoken languages?

In Harley & Ritter's (2002) review of results from the acquisition of pronominal systems in 6 languages as reported in 10 studies, they report acquisition ordering in support of their three proposed default features. In particular, they predict *uniformity* in the initial acquisition of 1st person singular or 3rd singular neuter/inanimate; acquisition of 2nd person after 1st person; and singular before plural. Our ASL data are in conformance with these predictions. We observed 3rd person singular inanimate in the first session for all participants; 1st person singular was the next to be acquired, and 2nd person singular was acquired after 1st person singular. We did not report children's FRU for plural pronouns, because three of the four participants used plural pronouns so infrequently that repeated use could not be determined, but for all three of them, their first instance of a plural was after the age of 2 (the exception is SAL, who achieved FRU for plurals at 1;08).

Overall, this pattern of acquisition is similar to that reported for spoken language pronouns and therefore we take the development of pointing signs as support for the analysis of (some of) them as pronouns.

C. Is there evidence for a distinction between 2nd and 3rd-person pointing signs? Does the time course of acquisition provide evidence for a developmental distinction between 2nd and 3rd person pointing signs in ASL?

Given the very low incidence of points to animate 3rd persons, we take with caution statistical analyses of the difference in FRU between 2nd and 3rd person signs. That said, this difference is significant in 2 of the 4 participants (ABY and JIL); for the other two children, 2nd and 3rd person signs either appeared in the same month (NED), or 3rd person signs appeared non-significantly later than 2nd person signs (SAL). We might take this evidence as tentative support for a grammatical distinction. We can also compare the participants' FRU for inanimate 3rd person signs and 2nd person signs. For ABY, these signs were both produced in the first session, so no difference could be assessed. For the other three participants, the inanimate 3rd person singular signs were produced significantly earlier than the earliest 2nd person signs (JIL: $p(\text{IX vs. IX(addr)}) = .03$; NED: $p(\text{IX vs. IX(addr)}) < .001$; SAL: $p(\text{IX vs. IX(addr)}) < .0001$). These results suggest that the 2nd/3rd person distinction is grammatically present in ASL, contrary to the proposals by Meier (1990) and others.

D. Should pointing signs directed to abstract loci be analyzed differently from other pointing signs? Do signing children acquire these two types of pointing differently?

As mentioned earlier, ASL allows pointing signs directed towards abstract loci established in the signing space, which are associated with non-present referents. In the period analyzed, we did not observe children using pointing signs directed at such abstract loci. In earlier research, these points to abstract loci are described as emerging only around the age of 4 or later in children's narratives (Bellugi, 1988; Bellugi et al., 1993; Hoffmeister, 1978; Loew, 1984). The fact that such points only develop much later supports the possibility that they might be analyzed differently from other pointing signs. Alternatively, there are additional requirements for the use of such signs which might take longer to develop, even if the signs themselves are analyzed uniformly. For example, in order to effectively use points to abstract loci, children must understand the concept of abstractly associating a referent with a location, and they must be able to maintain the same locus across multiple mentions of the same referent (for more discussion, see Bellugi, 1988; Loew, 1984).

8. Conclusions

In this study, we found that the pointing signs used in sign languages develop very differently in young children acquiring ASL as compared to the gestural pointing used by non-signing children. In particular, we observed almost no points to people in non-signing children; while points to self and addressee emerged in signing children between the ages of 17 and 27 months. These results lend support to the analysis of at least some pointing signs as personal pronouns. As Petitto (1987) explicitly argued, sign-exposed children learn to categorize pointing signs linguistically and treat them as pronouns, despite their overwhelming similarity to the gestural pointing used by non-signers. Our data set only begins at the age of 17-19 months, so we cannot contribute to the observations made by Petitto and others of a U-shaped trajectory for the

development of pointing to people in signing children, as this involves pointing produced between the ages of 12 and 24 months. Nevertheless, like others, we see an increase in the use of 1st person signs starting between the ages of 17 and 24 months, when our data collection begins.

It should be noted that although our dataset is much larger than those of previous studies, it is still relatively small, and only includes four children acquiring ASL. Analysis of additional data from children acquiring ASL and other sign languages would greatly strengthen the answers to the research questions raised here, as well as other questions regarding pointing signs.

We also note that our analysis included points produced by children in isolation as well as points produced with other signs. In a preliminary analysis of data from one child (JIL), we found that points to objects and locations are used together with other signs in early sentences from the first observation. Similarly, points to self are used in sentences when they first appear, at 1;10. However, points to addressee and non-addressed persons only appear in sentences much later than their appearance as isolated signs. Further analysis of pointing used in sentences is currently in preparation.

The current chapter is focused more on the emergence of points to people, and does not dive deeply into the characteristics of points to objects and locations. Lu (2021) conducted a study of the pointing produced by three of the same children analyzed here, at 4-month intervals, along with 10 non-signing, English-speaking children. Her focus was on the development of deixis in speech, gesture, and sign. Among other inquiries, her coding distinguished between literal and displaced referents, the latter of which includes non-literal points and points to absent objects. She found that such points emerge at 2;02 and continue to be used thereafter, albeit at a very low proportion of all pointing signs. Further investigation of the nature of such points may

be useful in considering linguistic analyses of demonstrative and locative signs in comparison to co-speech pointing.

In sum, we find the acquisitional evidence supports linguistic arguments regarding the personal pronominal status of ASL points to self and others. Our results are in conformance with the acquisitional predictions made by the model of Harley & Ritter (2002), suggesting that this model may be useful for analyzing the ASL pronominal system. We find partial acquisitional support for a grammatical distinction between 2nd and 3rd person forms, and suggestive evidence that points to abstract loci should receive a distinct analysis. In more general terms, we encourage consideration of language acquisition data as a factor that may contribute additional evidence for researchers considering controversial linguistic analyses.

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